

(12) **United States Patent**
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(10) **Patent No.: US 12,409,523 B1**
(45) **Date of Patent: Sep. 9, 2025**

(54) **PORTABLE WOODWORKING SPINDLE
FLUTING JIG AND METHOD OF USE**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 260 days.

(21) Appl. No.: **18/215,584**

(22) Filed: **Jun. 28, 2023**

Related U.S. Application Data

(60) Provisional application No. 63/443,507, filed on Feb.
6, 2023.

(51) **Int. Cl.**
B23Q 16/00 (2006.01)
B23Q 16/02 (2006.01)
B27C 5/10 (2006.01)

(52) **U.S. Cl.**
CPC **B23Q 16/001** (2013.01); **B23Q 16/02**
(2013.01); **B27C 5/10** (2013.01)

(58) **Field of Classification Search**
CPC .. B23Q 16/001; B23Q 16/02; B23Q 17/2233;
B23Q 17/2428; B27C 5/00; B27C 5/10;
B23B 49/00; B23B 3/065; B24B 23/028;
G05B 2219/45229

USPC 144/371
See application file for complete search history.

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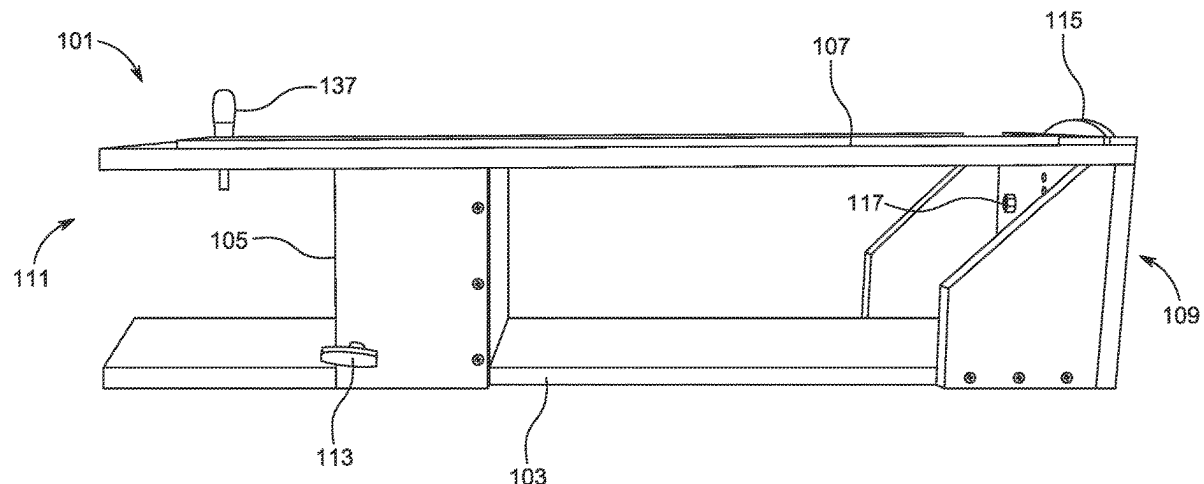
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(57) **ABSTRACT**

A spindle flute jig for woodworking is disclosed. The flute
jig is freestanding and independent of woodworking lathes.
The flute jig comprises a base, a carriage, an upper frame
positioned substantially parallel to the base, a drive end, a
pivot end, an adjustable plunge router stop, a fixed plunge
router stop, a tool storage section, and a marking block.

2 Claims, 9 Drawing Sheets



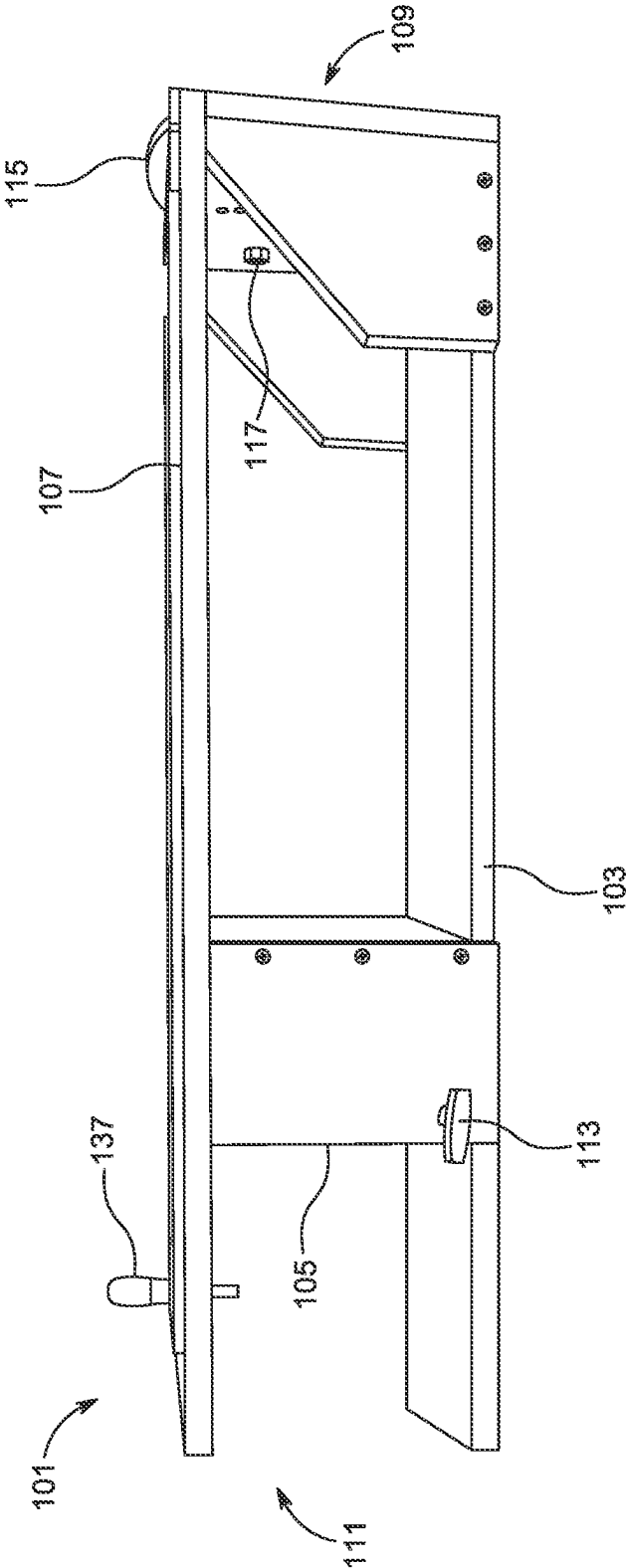


FIG. 1

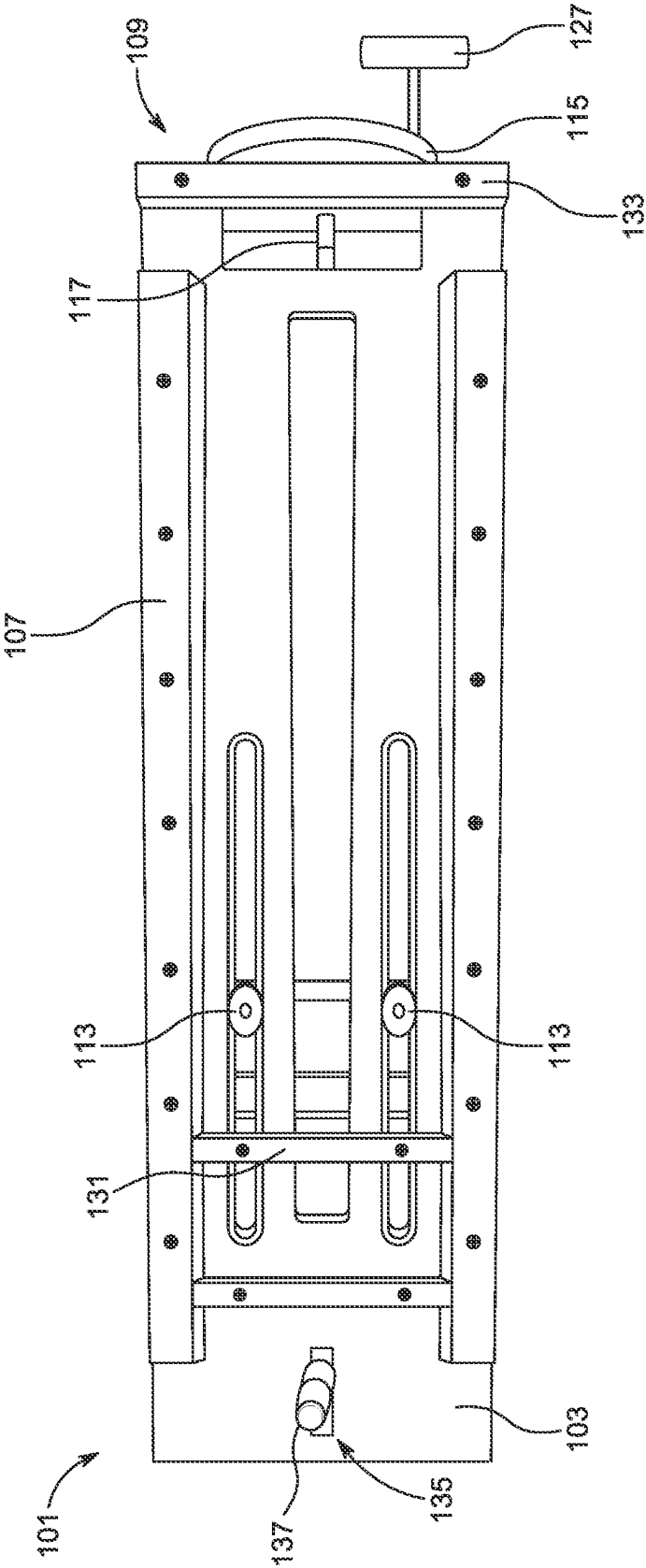


FIG. 2

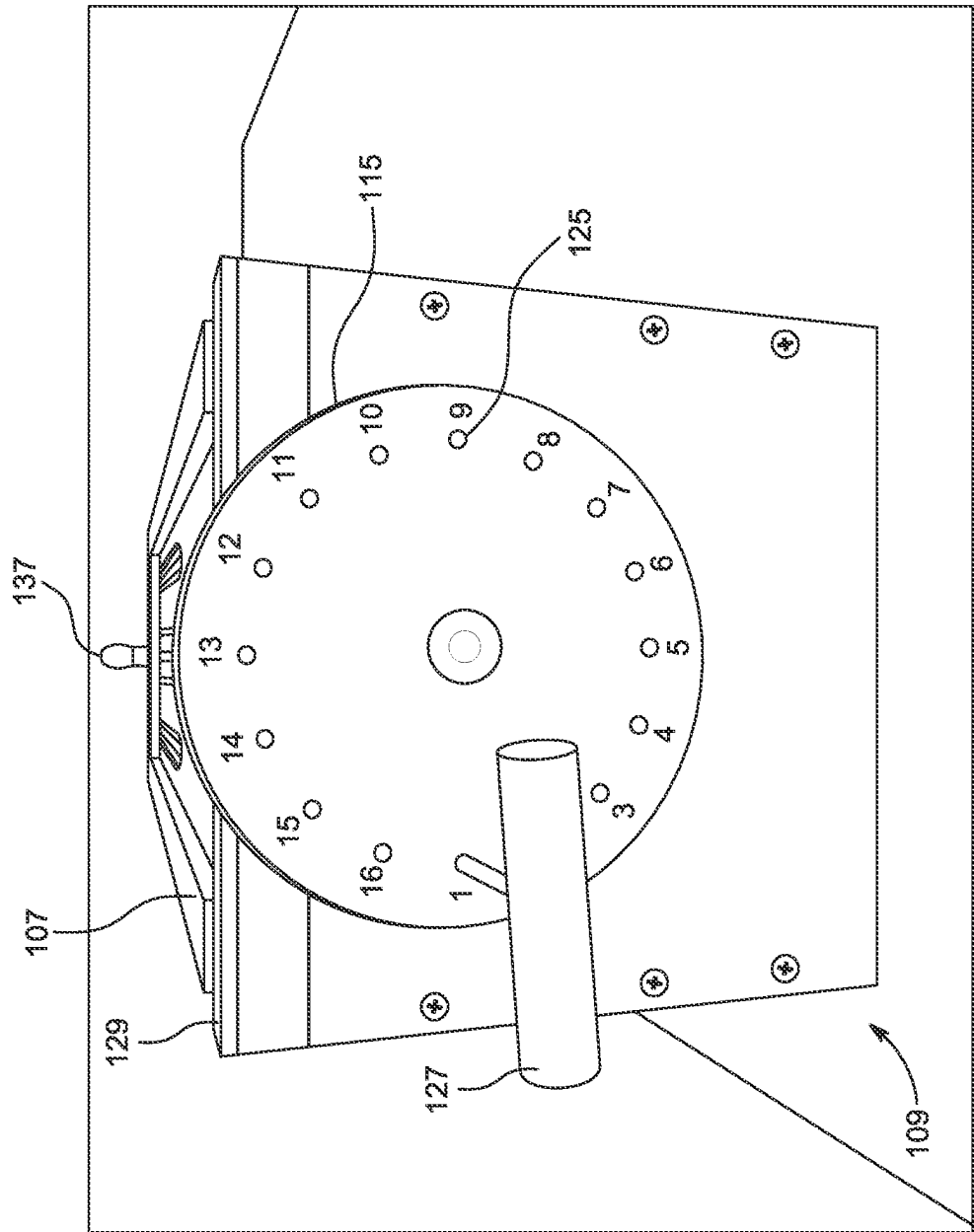


FIG. 3

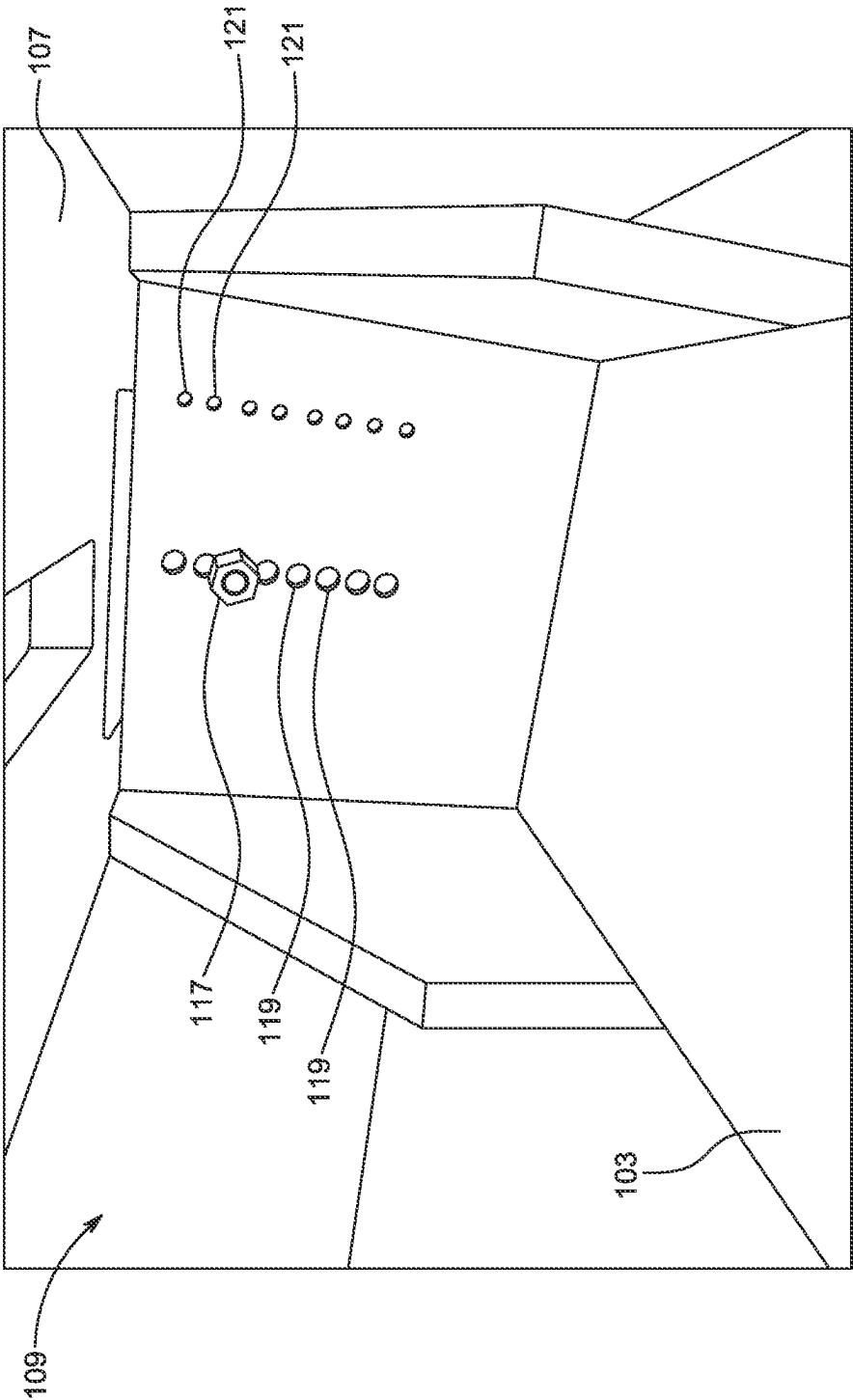


FIG. 4

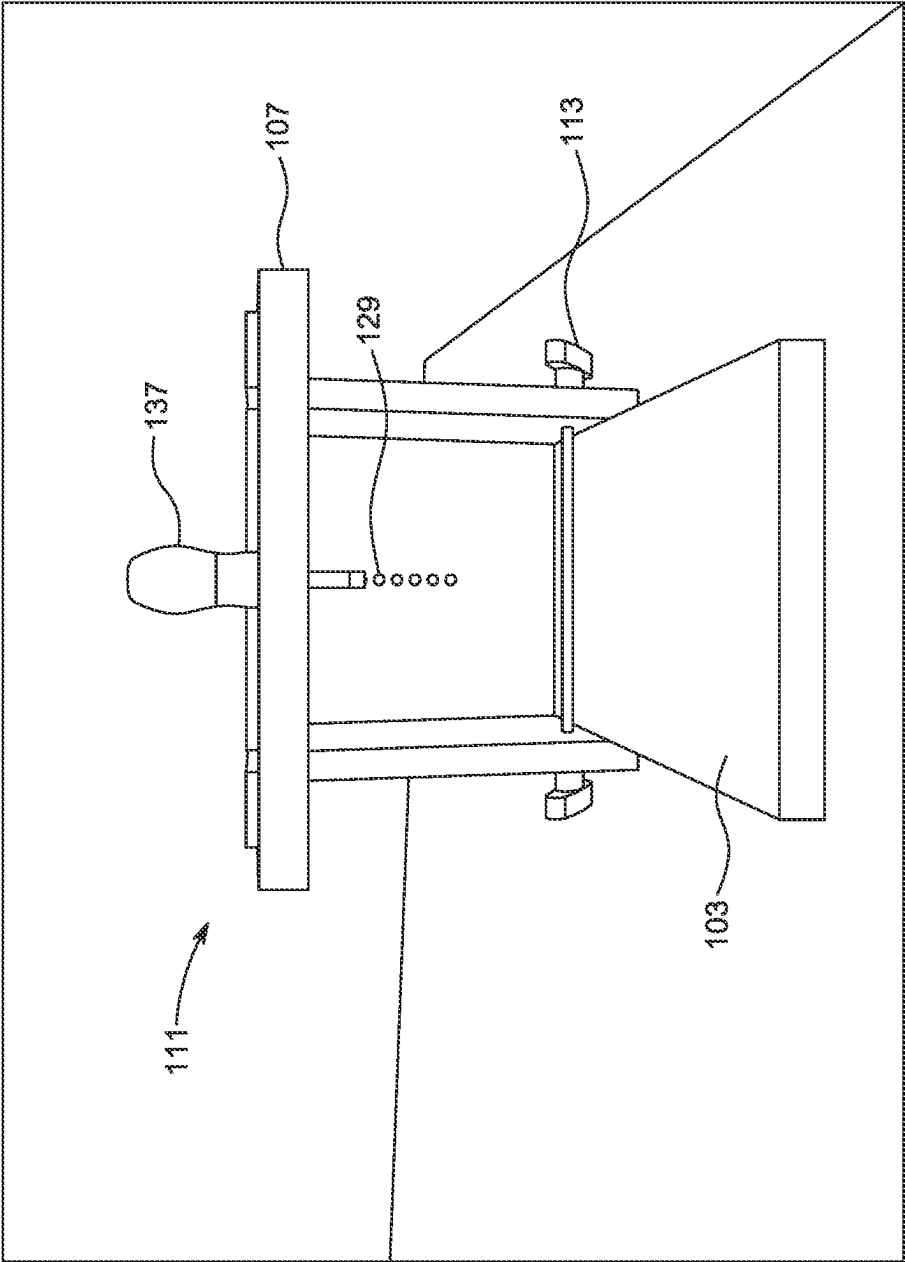


FIG. 5

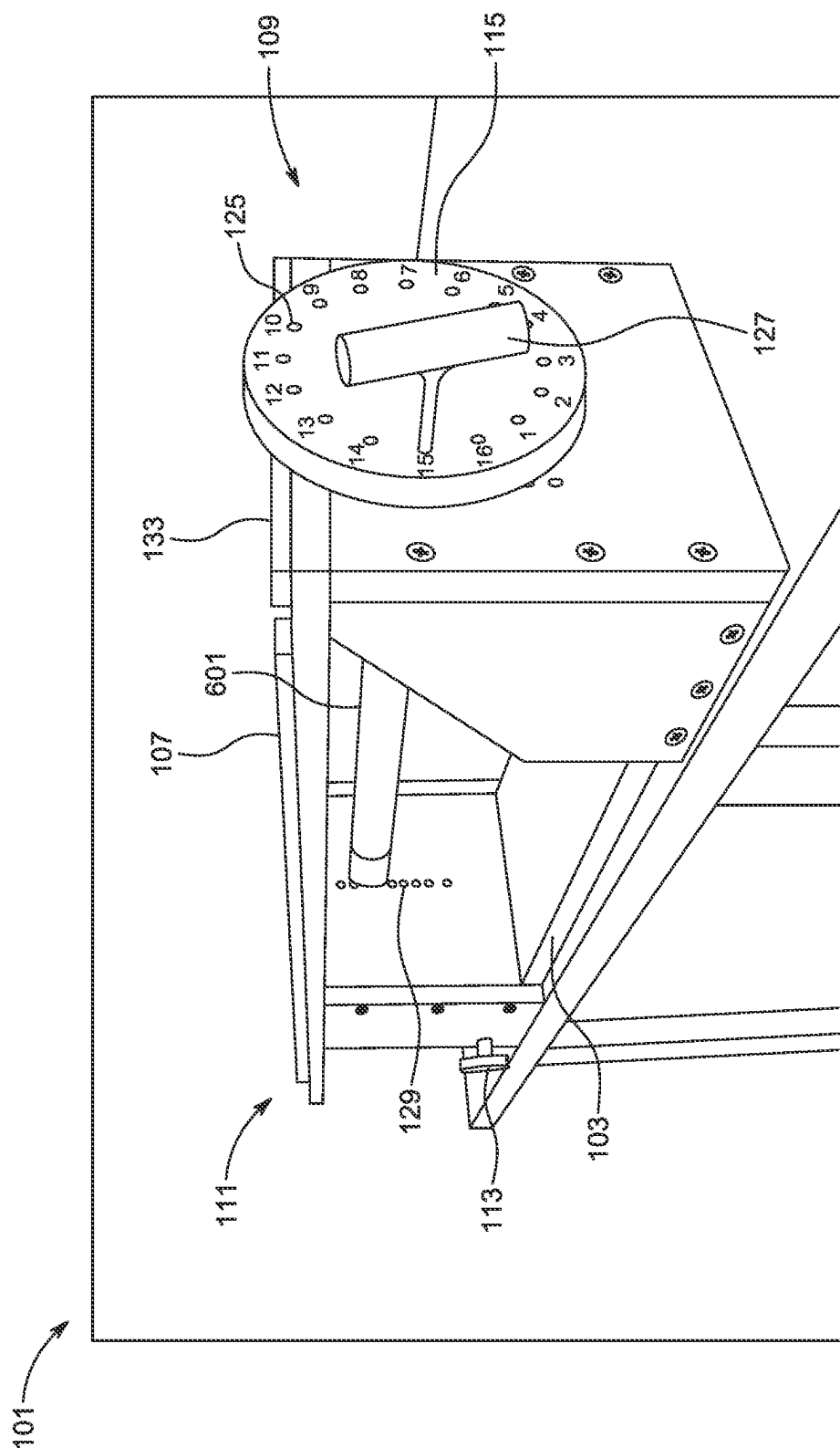


FIG. 6

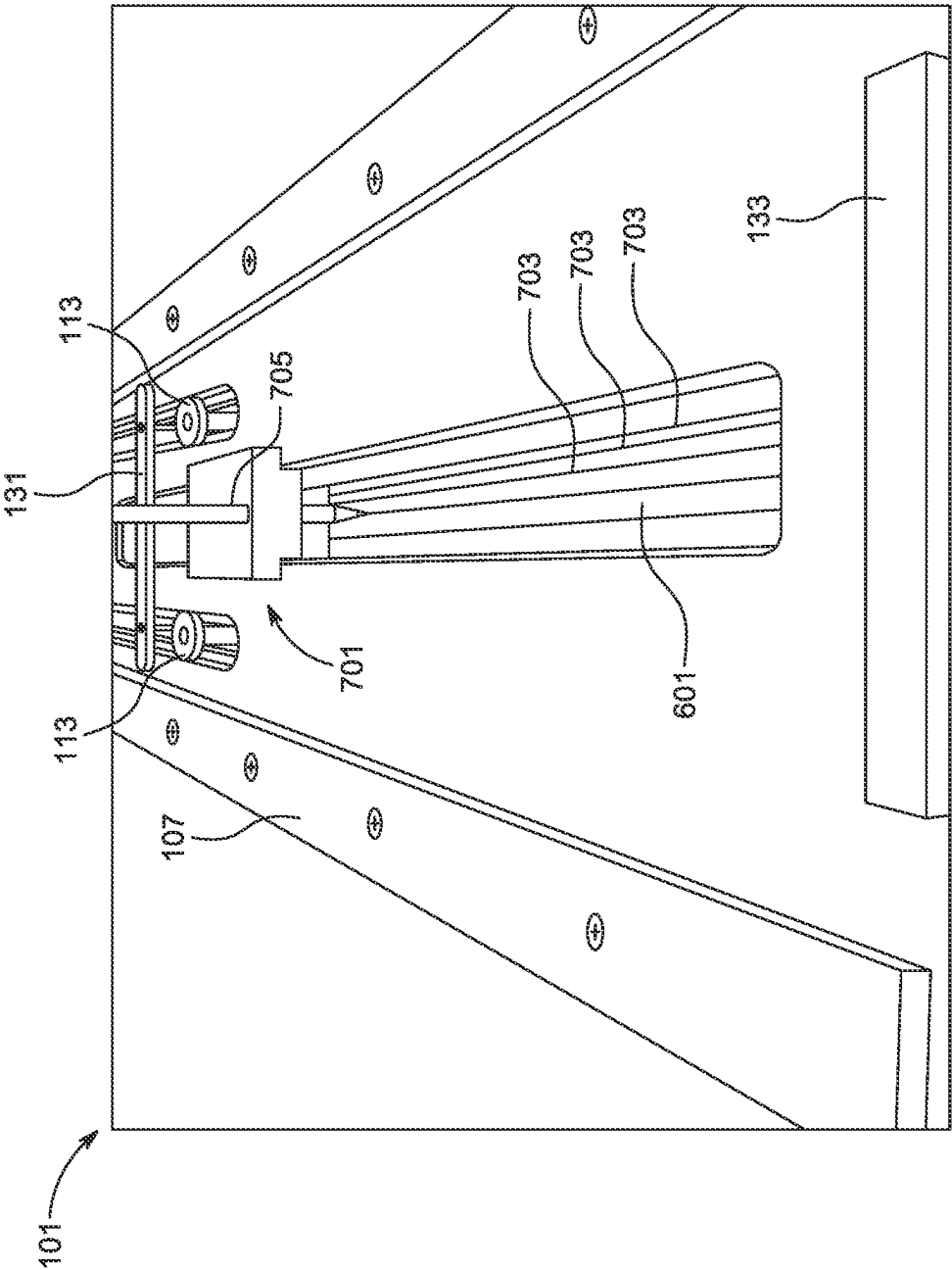


FIG. 7

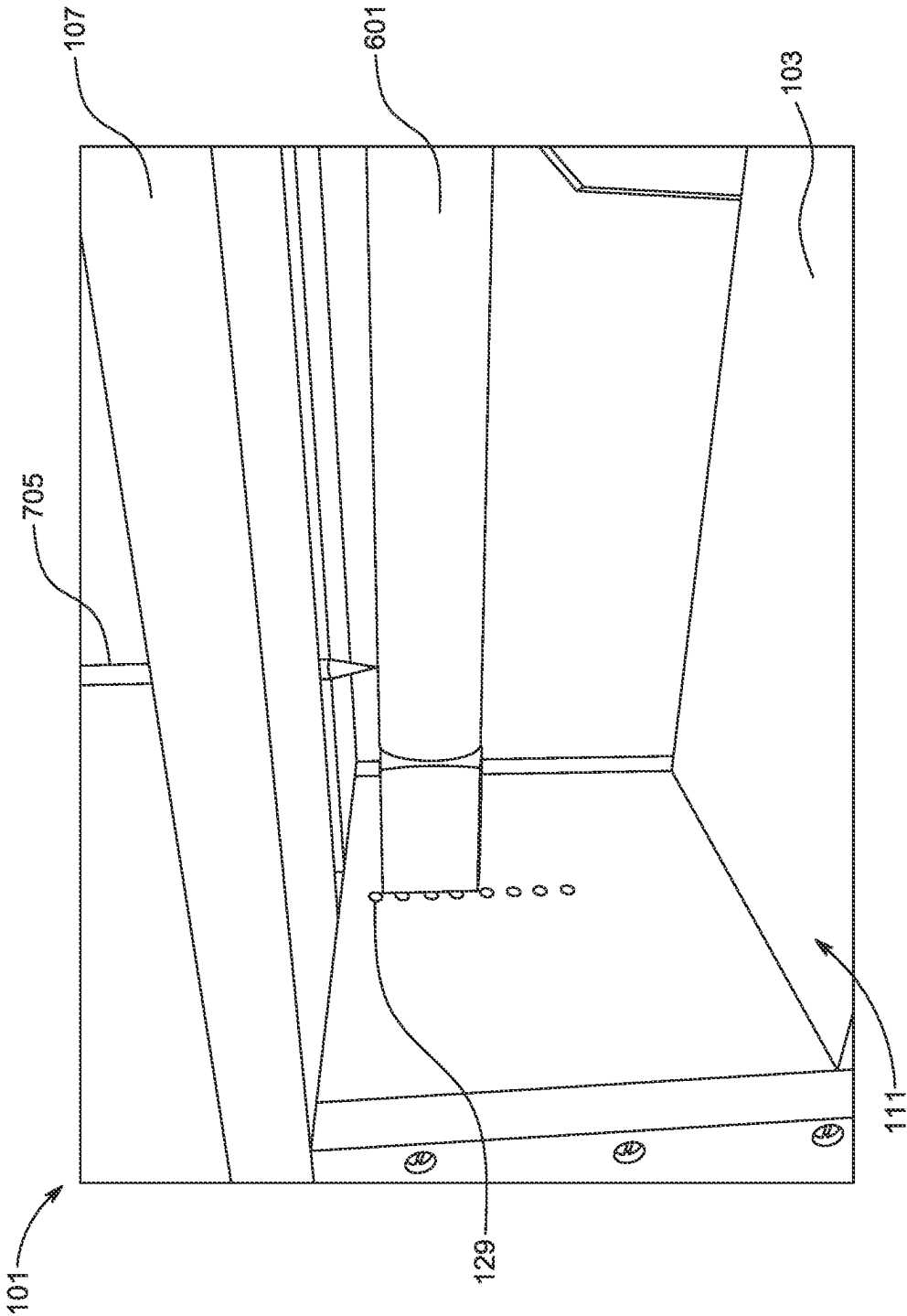


FIG. 8

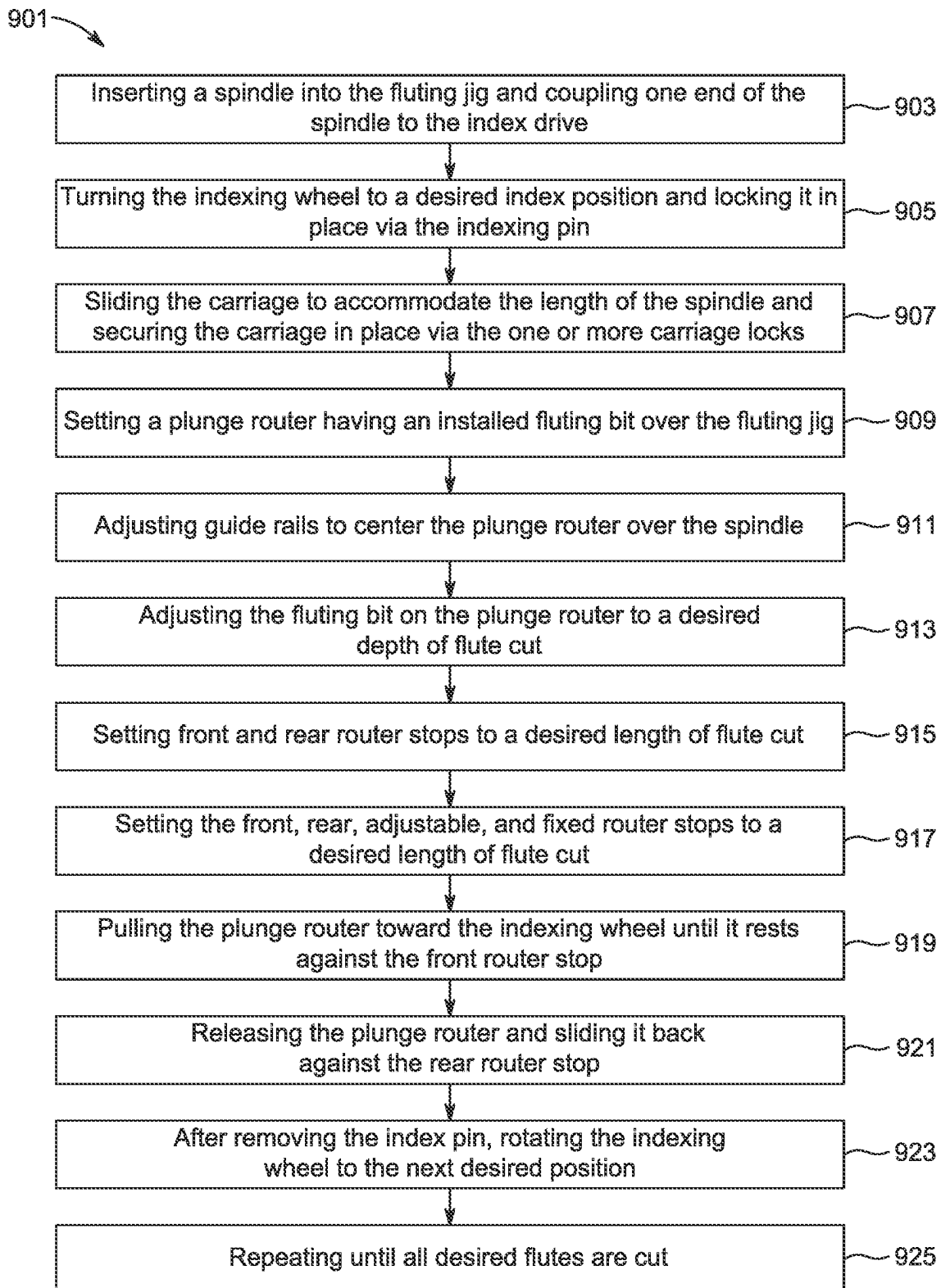


FIG. 9

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**PORTABLE WOODWORKING SPINDLE
FLUTING JIG AND METHOD OF USE****CROSS REFERENCE TO RELATED
APPLICATIONS**

This application claims the benefit of U.S. Provisional Application 63/443,507, filed Feb. 6, 2023, of which is hereby incorporated by reference in its entirety.

BACKGROUND**1. Field of the Invention**

The present invention relates generally to woodworking tools, and more specifically to a woodworking spindle fluting jig capable of allowing precise flute production on a wooden spindle.

2. Description of Related Art

Wooden spindles are often used to provide vertical support for stair railing, chairs, tables, and other types of furniture. Many wooden spindles incorporate decorative details to enhance their overall appearance such as flutes. Flutes are grooves that often run parallel to the spindle's long axis. Traditionally, flutes are cut by hand with a fluting chisel. This, however, requires a considerable amount of time and sophisticated carving skills to ensure equal spacing between each flute.

Accordingly, it is an object of the present invention to provide a portable woodworking spindle fluting jig that addresses the limitations discussed above.

DESCRIPTION OF THE DRAWINGS

The novel features believed characteristic of the embodiments of the present application are set forth in the appended claims. However, the embodiments themselves, as well as a preferred mode of use, and further objectives and advantages thereof, will best be understood by reference to the following detailed description when read in conjunction with the accompanying drawings, wherein:

FIG. 1 is a profile view of a woodworking spindle fluting jig in accordance with a preferred embodiment of the present invention;

FIG. 2 is a top view of the fluting jig of FIG. 1;

FIG. 3 is a front perspective view of the drive end of FIG. 1;

FIG. 4 is a closeup underside view of the drive end of FIG. 1;

FIG. 5 is a front perspective view of the pivot end of FIG. 1;

FIG. 6 is a side perspective view of the fluting jig of FIG. 1, illustrating a plunge router mounted thereon to perform fluting production on a spindle;

FIG. 7 is a top perspective view of the fluting jig of FIG. 6, illustrating a marking block;

FIG. 8 is a closeup underside view of the fluting jig of FIG. 6, illustrating equally spaced linear lines created by the marking block of FIG. 7; and

FIG. 9 is a flowchart of a method of use of the fluting jig of FIG. 1.

While the system and method of use of the present application is susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and are herein

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described in detail. It should be understood, however, that the description herein of specific embodiments is not intended to limit the invention to the particular embodiment disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present application as defined by the appended claims.

**DETAILED DESCRIPTION OF THE
PREFERRED EMBODIMENT**

Illustrative embodiments of the system and method of use of the present application are provided below. It will of course be appreciated that in the development of any actual embodiment, numerous implementation-specific decisions will be made to achieve the developer's specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.

The system and method of use in accordance with the present application overcomes one or more of the above-discussed problems commonly associated with conventional woodworking tools. Specifically, the present invention provides for a woodworking spindle fluting jig that is designed to hold a wooden spindle so that a cutting tool such as a plunge router can be employed to make flutes on the spindle with precision. These and other unique features of the system and method of use are discussed below and illustrated in the accompanying drawings.

The system and method of use will be understood, both as to its structure and operation, from the accompanying drawings, taken in conjunction with the accompanying description. Several embodiments of the system are presented herein. It should be understood that various components, parts, and features of the different embodiments may be combined together and/or interchanged with one another, all of which are within the scope of the present application, even though not all variations and particular embodiments are shown in the drawings. It should also be understood that the mixing and matching of features, elements, and/or functions between various embodiments is expressly contemplated herein so that one of ordinary skill in the art would appreciate from this disclosure that the features, elements, and/or functions of one embodiment may be incorporated into another embodiment as appropriate, unless described otherwise.

The preferred embodiment herein described is not intended to be exhaustive or to limit the invention to the precise form disclosed. It is chosen and described to explain the principles of the invention and its application and practical use to enable others skilled in the art to follow its teachings.

Referring now to the drawings wherein like reference characters identify corresponding or similar elements throughout the several views, FIG. 1 through 8 depict several views of a woodwork spindle fluting jig 101 in accordance with a preferred embodiment of the present application.

In the contemplated embodiment, the fluting jig 101 includes a base 103, a carriage 105, an upper frame 107, a drive end 109, and a pivot end 111. The upper frame 107 runs substantially parallel to the base 103 and is supported by the carriage 105 and the drive end 109.

The carriage **105** is configured to slide along the base **103** and the upper frame **107** to accommodate the length of a spindle **601**, and is locked into place via a plurality of carriage locks **113**.

The drive end **109** includes an indexing wheel **115** secured thereto via an index drive **117**. It should be understood that the index drive **117** corresponds with an end of a lathe drive (not shown). The drive end **109** also includes a plurality of indexing drive holes **119** and a plurality of indexing holes **121**. The indexing drive holes **119** allows a spindle of varying sizes to be accommodated within the fluting jig **101**. The indexing holes **121** are aligned with the indexing drive holes **119**. The index wheel **115** includes a plurality of indexing wheel holes **125** for enabling equally spaced lines on a turning. An indexing pin **127** is inserted into an indexing wheel hole **125** through an indexing hole **121** to secure the desired index in place.

The pivot end **111** includes a plurality of pivot adjustment holes **129** that align with the indexing drive holes **119**.

The fluting jig **101** also includes an adjustable plunge router stop **131** and a fixed plunge router stop **133**. The adjustable router stop **131** is positioned over the pivot end **111** and runs perpendicular across the upper frame **107**. The fixed router stop **133** is mounted over the drive end **109** and runs perpendicular across the upper frame **107**. During use, the base of a plunge router (not shown) makes contact with both the adjustable router stop **131** and the fixed router stop **133** at the opposing ends of the fluting jig **101**. The adjustable router stop **131** allows the user to adjust where flute carving begins from the end of a spindle. The fixed router stop **133** is configured to prevent the plunge router from dislodging off of the drive end **109**. It should be appreciated that the distance from the end of a spindle on the drive end **109** to the initial flute carving on the pivot end **111** can also be adjusted by placing a strip of wood of a predetermined width perpendicular across the top and in front of the fixed router stop **133**.

The fluting jig **101** also includes guide rails (not shown) configured to guide the base of the plunge router down the length of the fluting jig **101**. The guide rails are adjustable on each side of the fluting jig **101**. It should be appreciated that the guide rails are slotted at the fasteners to allow them to move toward or away from each other to accommodate various base diameters of plunge routers.

The fluting jig **101** also includes a front router stop **133** as shown in FIGS. 2 and 7. This router stop is fixed and prevents the router from being pulled accidentally from the jig. The fluting jig **101** also includes a read adjustable stop **131** as shown in FIGS. 2 and 7. This gives the woodworker the ability to adjust where the flutes will start from the end of the spindle.

The fluting jig **101** also includes a tool storage section **135** for storing various tools therein such as a screwdriver **137**.

The fluting jig **101** further includes a marking block **701** configured to mark equally spaced lines **703** along the length of and around the circumference of a spindle **601** via marking implement **705**, such as a pencil. The marking block **701** is inserted on top of the fluting jig **101** with the marking implement **705** resting on the spindle. The user can slide the marking block **701** from back to front, creating a line on the spindle. The user can also turn the indexing wheel **115** to the desired number of lines required around the spindle **601**. It should be appreciated that this allows the production of spiral-type of spindles or legs which can then be transferred to a woodworking lathe to complete.

It should be appreciated that the fluting jig can be used to cut a single flute on each side of a square spindle.

It should also be appreciated that one of the unique features believed characteristic of the present application is that the fluting jig **101** is freestanding and is independent from a woodworking lathe. Other similar jigs are mounted to a lathe with indexing capabilities. Spindles turned on lathes without the benefit of an indexing wheel can be inserted into the fluting jig **101** to cut the desired flutes. In addition, wood spindles can also be purchased and inserted into the fluting jig **101** for users without a woodworking lathe.

In FIG. 9, a flowchart **901** depicts a method of use of the fluting jig **101**. First, a spindle is inserted into the fluting jig with one end coupled to the index drive, as shown with box **903**. Next, the indexing wheel is turned to a desired index position and is locked in place via the indexing pin, as shown with box **905**. The carriage is slid to accommodate the length of the spindle and the carriage is secured in place via the one or more carriage locks, as shown with box **907**. A plunge router having an installed fluting bit can be set over the fluting jig, as shown with box **909**.

Guide rails are then adjusted to center the plunge router over the turned spindle, as shown with box **911**. The fluting bit is adjusted using the plunge stop feature on the router to the desired depth of the flute cut, as shown with box **913**. The front, rear, adjustable, and fixed router stops are set to the desired length of the flute cut, as shown with box **915**.

The plunge router is then set against the rear stop and is used to plunge into the desired depth, as shown with box **917**. Next, the plunge router is pulled toward the indexing wheel until it rests against the front router stop, as shown with box **919**. The plunge router is then released and slid back against the rear router stop, as shown with box **921**. After removing the indexing pin, the indexing wheel is rotated to the next desired position and secured with the indexing pin, as shown with box **923**. The user can repeat these steps until all desired flutes are cut, as shown with box **925**.

It should be understood that before cutting flutes into a spindle, the number of flutes desired is based on the diameter of the spindle. For example, for larger diameter spindles, as many as 16 flutes evenly spaced around the turned spindle can be cut. In another example, for smaller diameter spindles, every other indexing hole can be skipped.

The particular embodiments disclosed above are illustrative only, as the embodiments may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. It is therefore evident that the particular embodiments disclosed above may be altered or modified, and all such variations are considered within the scope and spirit of the application. Accordingly, the protection sought herein is as set forth in the description. Although the present embodiments are shown above, they are not limited to just these embodiments, but are amenable to various changes and modifications without departing from the spirit thereof.

What is claimed is:

1. A spindle fluting jig for woodworking, comprising:
 - a base;
 - a carriage;
 - an upper frame positioned substantially parallel to the base;
 - a drive end, the drive end having:
 - an index wheel secured thereto via an index drive;
 - a plurality of indexing drive holes; and
 - a plurality of indexing holes;
 - wherein the plurality of indexing drive holes and the plurality of indexing holes are aligned therebetween;

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a pivot end, the pivot end having:
 a plurality of pivot adjustment holes aligned with the plurality of indexing drive holes;
 an adjustable plunge router stop positioned over the pivot end and extended perpendicularly across the upper frame, the adjustable plunge router stop configured to allow the user to adjust where flute carving begins from an end of a spindle;
 a fixed plunge router stop mounted over the drive end and extended perpendicularly across the upper frame, the fixed plunge router stop configured to prevent a plunge router from dislodging off of the drive end;
 a fixed front plunge router stop that prevents the router from being pulled accidentally from the jig;
 an adjustable rear plunge router stop that gives the woodworker the ability to adjust where the flutes will start from the end of the spindle;
 a pair of guide rails configured to guide a base of a plunge router down a length of the fluting jig and configured to accommodate one or more base diameters of a plunge router;
 a tool storage section for storing one or more tools therein; and
 a marking block configured to mark a plurality of equally spaced lines along a length of and around a circumference of a spindle via marking implement;
 wherein the upper frame runs substantially parallel to the base and is supported by the carriage and the drive end;
 wherein the carriage is configured to slide along the base and the upper frame to accommodate a length of a spindle and is locked into place via a plurality of carriage locks.

2. A method of producing flutes on a spindle, the method comprising:
 providing a spindle fluting jig, the spindle fluting jig comprising:
 a base;
 a carriage;
 an upper frame positioned substantially parallel to the base;
 a drive end, the drive end having:
 an index wheel secured thereto via an index drive;
 a plurality of indexing drive holes; and
 a plurality of indexing holes;
 wherein the plurality of indexing drive holes and the plurality of indexing holes are aligned therebetween;
 a pivot end, the pivot end having:
 a plurality of pivot adjustment holes aligned with the plurality of indexing drive holes;
 an adjustable plunge router stop positioned over the pivot end and extended perpendicularly across the upper frame, the adjustable plunge router stop configured to allow the user to adjust where flute carving begins from an end of a spindle;

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a fixed plunge router stop mounted over the drive end and extended perpendicularly across the upper frame, the fixed plunge router stop configured to prevent a plunge router from dislodging off of the drive end;
 a fixed front plunge router stop that prevents the router from being pulled accidentally from the jig;
 an adjustable rear plunge router stop that gives the woodworker the ability to adjust where the flutes will start from the end of the spindle;
 a pair of guide rails configured to guide a base of a plunge router down a length of the fluting jig and configured to accommodate one or more base diameters of a plunge router;
 a tool storage section for storing one or more tools therein; and
 a marking block configured to mark a plurality of equally spaced lines along a length of and around a circumference of a spindle via marking implement;
 wherein the upper frame runs substantially parallel to the base and is supported by the carriage and the drive end;
 wherein the carriage is configured to slide along the base and the upper frame to accommodate a length of a spindle and is locked into place via a plurality of carriage locks;
 inserting a spindle into the spindle fluting jig and coupling one end of the spindle to the index drive;
 turning the index wheel to a desired index position and locking said desired index position via the indexing pin;
 sliding the carriage to accommodate a length of the spindle;
 securing the carriage in place via the one or more carriage locks;
 setting a plunge router having an installed fluting bit over the spindle fluting jig;
 adjusting the guide rails to center the plunge router over the spindle;
 adjusting the fluting bit on the plunge router to a desired depth of flute cut;
 setting the front, rear, adjustable, and fixed router stops to a desired length of the flute cut;
 setting the plunge router against the rear stop and using said plunge router to plunge into the desired depth;
 pulling the plunge router toward the index wheel until said plunge router rests against the front router stop;
 releasing the plunge router and sliding said plunge router against the rear router stop;
 after removing the index pin, rotating the indexing wheel to another desired position; and
 repeating until all desired flutes are cut.

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